79

79

construction workers get injured in Canada every single day.

AWCBC · accepted lost-time claims, construction, 2021

natHacks 2026 · construction & manufacturing

Argus

why close calls get ignored

paperwork

blame

nothing happens

Aulin & Linderbäck, Lund University (2014) · Human Focus

a retrofit for any hard hat.

line of fire

rear sensors + buzzer

falls

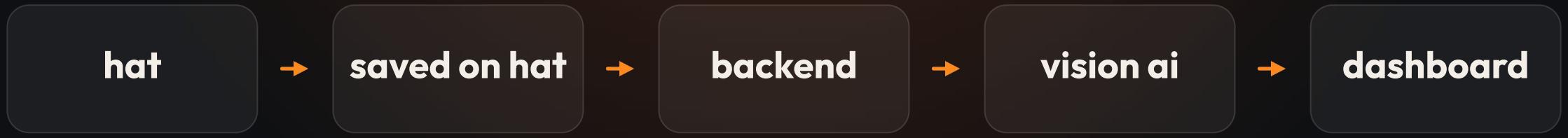
motion sensor

safety report

dash cam

plus a time-of-flight sensor that checks the hat is actually being worn, There is also a carbon dioxide, humidity, and temperature sensor aboard the retrofit to get a better understanding of environmental conditions at the worker level.

how it works on a site



let's show you.

to protect a crew, others need

~~a tag on every truck.~~

~~cameras on every pole.~~

~~a new helmet for every worker.~~

argus needs the hat they already own.

the retrofit gets us on site. the history keeps us there.

workers wear it because it isn’t watching them.

59% of construction laborers wouldn’t wear tracking wearables

EHS Today, 2025

what's next

$600M a year: everyone in canada who wears a hard hat

- train our own model on our data
- radar for vehicles
- certified for oil & gas zones

CAD · ~2M workers × $25/hat/month · StatCan 2025–26

the warning signs get ignored.
argus makes sure someone acts on them.

our inspiration · chelsea, quebec · 2023

the warning came two weeks early.

two weeks before

a reversing dump truck hits a parked roller. nobody is hurt. nothing changes.

september 11, 2023

a reversing dump truck kills a worker refuelling a bulldozer behind it.

CNESST investigation report, Construction FGK inc. (2023), p. 41

business model

per hat

$25/mo

per site

$499/mo

start

free pilot → paid

one prevented injury pays for 40 hats for a year.

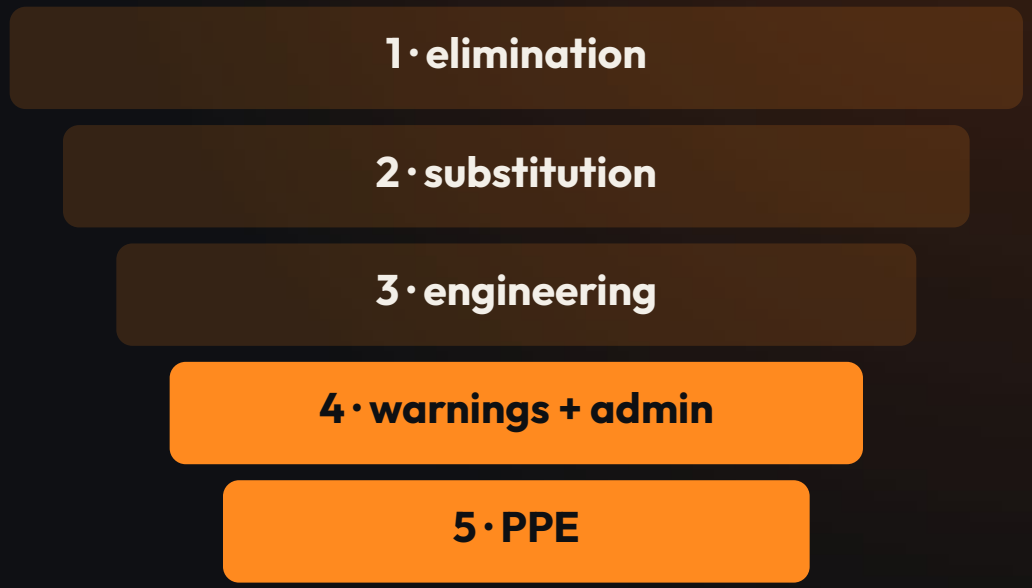
CAD · avg lost-time claim $12,219, WCB Alberta 2024

why we built it this way

clip-on retrofit	over	a new helmet
event-only clips	over	constant video
haptic buzz	over	sound or lights
ultrasonic + IMU	over	camera arrays
code scores risk	over	AI scores risk

hierarchy of controls

every report points up the pyramid.



report suggests: **remove the hazard**

report suggests: **change the method**

report suggests: **barriers, exclusion zones**

argus does: **the buzz + the report**

argus is: **worn on the hard hat**

NIOSH hierarchy of controls · most effective at the top

why this works

buyer

safety managers

proof

2 of us worked on sites

economics

$220 parts → $25/mo

why now

AI writes the report

our market



TAM · every hard-hat worker in canada

$600M / yr

~2M workers × $300 a year

SAM · alberta construction

$75M / yr

~250K workers × $300 a year

SOM · 3-year goal

$750K / yr

2,500 hats, 1% of alberta

CAD · $300 = $25/hat/month × 12 · StatCan 2025–26 · Job Bank Alberta 2025

competitors

	Form	Warns wearer	Writes report	Records	Price
Argus	Clip-on, any hat	Yes, haptic	Yes, AI	Event clips only	CAD 25/hat/mo
SenseHuman AI	Own 4-camera helmet	Yes, directional	Not public	Video, no audio	Not public
Guardhat (Aatmunn)	Own helmet	Yes	Not public	Live location + sensors	From US$1,500
Leica PA10	Tags + machine anchors	Yes, tag	Incident log (add-on)	Proximity events	Not public
PIP + Quin hard hat	Own helmet	No, alerts after a fall	Not public	Impacts, falls	Not public
NEC excavator AI	Cameras on machines	Not stated	Yes, gen-AI	Machine video	Launch FY2027/28
Forms / apps (e.g. HCSS)	Phone or paper	No	By hand	Photos	Staff time

others warn or report. argus does both, from the hat a crew already owns.

From public materials. “Not public” ≠ they can’t.

backup b · detection logic

how detection and scoring work

Event	Trigger	Guard against false alarms
Line of fire	Object within 2 m behind you, closing faster than 1.5 m/s, contact in under 1.3 s	Subtracts wearer's own motion; 2 sweeps to confirm; 5 s cooldown
Fall	Free-fall (under 0.35 g, 200+ ms), then over 3 g within 1 s	200 ms minimum rules out bumps; 5 s cooldown
Impact / hat off	Impact: spike over 6 g. Hat off: no head seen for 3 s	Removed: logged only. Knocked off: clip + report
Report score	AI rates likelihood and severity 1–5; code multiplies	Schema check, up to 3 tries; no usable video = sensor-only report

why the camera isn’t collecting data

we asked

our 2 teammates who worked on sites wouldn’t wear a hat that watches them

so we built

clips only around a close call, aimed at the hazard, not the worker

because

a hat nobody wears collects nothing

59% of construction laborers wouldn’t wear tracking wearables · EHS Today, 2025

where the $25 goes

one hat · retail parts today

~$220

raspberry pi 4 (1 GB)	$52
camera	$35
battery pack	$30
distance + motion sensors	$29
wiring + printed mount	$20
buzzers + driver	$12
memory card	$12
assembly	$30

goal at volume: ~$160

per hat · per year · $300 in	today	at volume
hardware, over 2 years	$110	$80
support	$24	$24
replacing lost + broken hats	$23	$23
cloud + AI reports	$20	$12
replacement batteries	$12	$12
card payment fees	$9	$9
total cost	$198	$160
gross margin	**34%**	**47%**

at volume = bulk parts + one AI model

CAD · retail midpoints · support, hat life and loss rate are our assumptions